

CMP513 - Computer Graphics and Image Processing

Tutorial Activity

A. Topic: Properties of matrix inverse

1. $(AB)^{-1} = B^{-1}A^{-1}$
2. $(A^{-1})^{-1} = A$
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(kA)^{-1} = \frac{1}{k}A^{-1}$

Prove these properties for given values of A and B.

Property 1: $(AB)^{-1} = B^{-1}A^{-1}$

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$$

Property 2: $(A^{-1})^{-1} = A$

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

Property 3: $(A^T)^{-1} = (A^{-1})^T$

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

Property 4: $(kA)^{-1} = \frac{1}{k}A^{-1}$

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \quad \text{consider } k = 5.$$

B. Find out the applications of matrix inverse in computer graphics and image processing.